

考研竞赛凯哥 第5次周考

考试时间: 4月10日上午 08:30~10:00

讲解时间: 4月10日上午 10:00~12:00

注1: 本次周考的范围为《极限与连续》《一元微分的计算与证明》《不定积分与定积分计算》

注2: 周考, 每周一次, 比较频繁, 所以题量减半, 考试时间减半, 且选填题与解答题的比例随机;

月考, 每月一次, 完全按照考研的标准进行设置.

注3: 无论周考还是月考, 目的都是查漏补缺和巩固所学知识, 所以题目的难度至少持平或略高于真题的平均难度, 大家一定不要因为模考成绩不理想而灰心, 考前犯的任何错误都应该珍惜!

$$1. \lim_{x \rightarrow 0} \left[\frac{\int_0^x e^{t^2} \cos t dt}{\ln^2(1+x)} - \frac{1}{e^x - 1} \right]$$

$$\text{解: } I = \lim_{x \rightarrow 0} \frac{(e^x - 1) \int_0^x e^{t^2} \cos t dt - \ln^2(1+x)}{(e^x - 1) \ln^2(1+x)} = \lim_{x \rightarrow 0} \frac{(e^x - 1) \int_0^x e^{t^2} \cos t dt - \ln^2(1+x)}{x^3}$$

$$x \rightarrow 0 \text{ 时, } e^{x^2} \cos x = [1 + x^2 + o(x^2)] \left[1 - \frac{1}{2}x^2 + o(x^2) \right] = 1 + \frac{1}{2}x^2 + o(x^2)$$

$$\text{故 } \int_0^x e^{t^2} \cos t dt = x + \frac{1}{6}x^3 + o(x^3).$$

$$\text{故 } I = \lim_{x \rightarrow 0} \frac{\left[x + \frac{x^2}{2} + o(x^2) \right] \left[x + \frac{1}{6}x^3 + o(x^3) \right] - \left[x - \frac{1}{2}x^2 + o(x^2) \right]^2}{x^3}$$

$$= \lim_{x \rightarrow 0} \frac{(1-1)x^2 + \left[\frac{1}{2} - 2 \cdot 1 \cdot \left(-\frac{1}{2} \right) \right] x^3 + o(x^3)}{x^3} = \frac{1}{2} - 2 \cdot 1 \cdot \left(-\frac{1}{2} \right) = \frac{3}{2}.$$

$$2. \text{ 设 } f(x) \text{ 满足 } f'(a) \neq 0, f''(a) \text{ 存在, 求 } \lim_{x \rightarrow a} \left(\frac{1}{f(x) - f(a)} - \frac{1}{(x-a)f'(a)} \right)$$

$$\text{解: } I = \lim_{x \rightarrow a} \left(\frac{1}{f(x) - f(a)} - \frac{1}{(x-a)f'(a)} \right) = \lim_{x \rightarrow a} \frac{(x-a)f'(a) - [f(x) - f(a)]}{[f(x) - f(a)](x-a)f'(a)}$$

$$= \lim_{x \rightarrow a} \frac{(x-a)f'(a) - \left[f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + o((x-a)^2) - f(a) \right]}{[f(x) - f(a)](x-a)f'(a)}$$

$$= \lim_{x \rightarrow a} \frac{-\frac{f''(a)}{2!}(x-a)^2}{[f(x) - f(a)](x-a)f'(a)} = \lim_{x \rightarrow a} \frac{-\frac{f''(a)}{2!}}{\frac{f(x) - f(a)}{x-a} f'(a)} = \lim_{x \rightarrow a} \frac{-\frac{f''(a)}{2!}}{[f'(a)]^2} = -\frac{1}{2} \frac{f''(a)}{[f'(a)]^2}$$

8. 设 $x_1 > 0$, $x_{n+1} = \frac{3(1+x_n)}{3+x_n}$, 证明 $\lim_{n \rightarrow \infty} x_n$ 存在并求该极限值.

法 1: 万能模板, 也是凯哥全程班里一直强调的方法——

(草稿纸上: $A = \frac{3(1+A)}{3+A} \Rightarrow A^2 = 3$, 结合 $A \geq 0$ 得 $A = \sqrt{3}$, 故需讨论 x_1 与 $\sqrt{3}$ 的大小关系)

(1) 若 $x_1 = \sqrt{3}$, 则 $x_2 = \sqrt{3}$ 、 $x_3 = \sqrt{3}$ 、 $\dots$, 即 $\{x_n\}$ 恒为 $\sqrt{3}$, 故 $\lim_{n \rightarrow \infty} x_n = \sqrt{3}$;

(2) 若 $x_1 > \sqrt{3}$, (此时猜测 $\{x_n\}$ 单调递减趋向于 $\sqrt{3}$, 故 $\sqrt{3}$ 是最精确的下界).

1) 有界性:

① $x_1 > \sqrt{3}$;

② 假设 $x_n > \sqrt{3}$;

③ $x_{n+1} = \frac{3(1+x_n)}{3+x_n}$, 令 $f(x) = \frac{3(1+x)}{3+x} = \frac{3(3+x)-6}{3+x} = 3 - \frac{6}{x+3}$, 显然 $f(x)$ 在 $x > 0$ 时递增,

故 $x_{n+1} = f(x_n) > f(\sqrt{3}) = \sqrt{3}$.

由数学归纳法可得, $x_n > \sqrt{3}$ 恒成立, 即 $\{x_n\}$ 有下界.

3) 单调性:

$x_{n+1} - x_n = \frac{3(1+x_n)}{3+x_n} - x_n = \frac{3(1+x_n) - x_n(3+x_n)}{3+x_n} = \frac{3-x_n^2}{3+x_n} < 0$, 故 $\{x_n\}$ 单调递减.

由单调有界准则可得, $\{x_n\}$ 收敛. 假设 $\lim_{n \rightarrow \infty} x_n = A$, 则 $A = \frac{3(1+A)}{3+A}$, 由显然 $A \geq 0$, 故 $A = \sqrt{3}$.

故当 $x_1 > \sqrt{3}$ 时, 也有 $\lim_{n \rightarrow \infty} x_n = \sqrt{3}$.

(3) 若 $0 < x_1 < \sqrt{3}$, 同(2)可证 $\{x_n\}$ 单调递增趋向于 $\sqrt{3}$, 即 $\lim_{n \rightarrow \infty} x_n = \sqrt{3}$.

综上, 无论 x_1 的取值如何, 恒有 $\lim_{n \rightarrow \infty} x_n = \sqrt{3}$.

法 2: 显然 $x_n > 0$, 对 x_{n+1} 作简单放缩—— $1 = \frac{3+x_n}{3+x_n} < \frac{3+3x_n}{3+x_n} = \frac{3(1+x_n)}{3+x_n} < \frac{3(3+x_n)}{3+x_n} = 3$,

即 $1 < x_{n+1} < 3$, 即 $n \geq 2$ 时恒有 $1 < x_n < 3$, 故 $\{x_n\}$ 既有上界也有下界.

令 $f(x) = \frac{3(1+x)}{3+x} = \frac{3(3+x)-6}{3+x} = 3 - \frac{6}{x+3}$, 显然 $f(x)$ 在 $x > 0$ 时递增,

这说明满足 $x_{n+1} = f(x_n)$ 的递推数列 $\{x_n\}$ 一定单调.

故由单调有界准则可得, $\{x_n\}$ 收敛. 假设 $\lim_{n \rightarrow \infty} x_n = A$, 则 $A = \frac{3(1+A)}{3+A}$, 由显然 $A \geq 0$, 故 $A = \sqrt{3}$.

法 3: 显然 $x_n > 0$, 令 $f(x) = \frac{3(1+x)}{3+x} = \frac{3(3+x)-6}{3+x} = 3 - \frac{6}{x+3}$, 其中 $x > 0$, 故 $|f'(x)| = \frac{6}{(x+3)^2} < \frac{6}{9}$.

故 $0 \leq |x_{n+1} - \sqrt{3}| = |f(x_n) - f(\sqrt{3})| = |f'(\xi)(x_n - \sqrt{3})| < \frac{6}{9} |x_n - \sqrt{3}|$, 一直迭代下去——

$$0 \leq |x_{n+1} - \sqrt{3}| < \frac{6}{9} |x_n - \sqrt{3}| < \left(\frac{6}{9}\right)^2 |x_{n-1} - \sqrt{3}| < \left(\frac{6}{9}\right)^3 |x_{n-2} - \sqrt{3}| < \cdots < \left(\frac{6}{9}\right)^n |x_1 - \sqrt{3}| \xrightarrow{n \rightarrow \infty} 0.$$

故由夹逼准则 $\lim_{n \rightarrow \infty} (x_{n+1} - \sqrt{3}) = 0$, 即 $\lim_{n \rightarrow \infty} x_n = \sqrt{3}$.

4. 设 $f(x) = \ln \frac{1+x}{5-x}$, 求 $f^{(100)}(3)$.

解: 在直播讲解中, 我们用的“找规律法”, 现在我们展示“泰勒展开对比系数法”

$$\begin{aligned} f(x) &= \ln \frac{1+x}{5-x} = \ln(4+x-3) - \ln(2+3-x) = \ln 4 + \ln\left(1 + \frac{x-3}{4}\right) - \ln 2 - \ln\left(1 - \frac{x-3}{2}\right) \\ &= \ln 2 + \sum_{n=1}^{\infty} (-1)^{n-1} \frac{\left(\frac{x-3}{4}\right)^n}{n} + \sum_{n=1}^{\infty} \frac{\left(\frac{x-3}{2}\right)^n}{n}, \end{aligned}$$

又由于 $f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(3)}{n!} (x-3)^n$, 对比 $(x-3)^n$ 前的系数可得 $\frac{1}{n} \left[\frac{(-1)^{n-1}}{4^n} + \frac{1}{2^n} \right] = \frac{f^{(n)}(3)}{n!}$,

解得 $f^{(n)}(3) = (n-1)! \left[\frac{(-1)^{n-1}}{4^n} + \frac{1}{2^n} \right]$.

5. $\int \ln(\sqrt{1+x} + \sqrt{1-x}) dx$

解: $I = \int \ln(\sqrt{1+x} + \sqrt{1-x}) dx = x \ln(\sqrt{1+x} + \sqrt{1-x}) - \int x \frac{\frac{1}{2\sqrt{1+x}} - \frac{1}{2\sqrt{1-x}}}{\sqrt{1+x} + \sqrt{1-x}} dx.$

这里我们单独化简等式右端的被积函数——

$$\begin{aligned} x \frac{\frac{1}{2\sqrt{1+x}} - \frac{1}{2\sqrt{1-x}}}{\sqrt{1+x} + \sqrt{1-x}} &\stackrel{\text{分子通分}}{=} \frac{1}{2} x \frac{\frac{\sqrt{1-x} - \sqrt{1+x}}{\sqrt{1-x^2}}}{\sqrt{1+x} + \sqrt{1-x}} = \frac{1}{2} \frac{x}{\sqrt{1-x^2}} \frac{\sqrt{1-x} - \sqrt{1+x}}{\sqrt{1+x} + \sqrt{1-x}} \\ &\stackrel{\text{分母有理化}}{=} \frac{1}{2} \frac{x}{\sqrt{1-x^2}} \frac{[\sqrt{1-x} - \sqrt{1+x}]^2}{(1+x) - (1-x)} = \frac{1}{2} \frac{x}{\sqrt{1-x^2}} \frac{(1-x) + (1+x) - 2\sqrt{1-x^2}}{2x} \\ &= \frac{1}{2} \frac{x}{\sqrt{1-x^2}} \frac{2 - 2\sqrt{1-x^2}}{2x} = \frac{1}{2} \frac{1}{\sqrt{1-x^2}} \frac{1 - \sqrt{1-x^2}}{1} = \frac{1}{2} \left(\frac{1}{\sqrt{1-x^2}} - 1 \right). \end{aligned}$$

$$I = x \ln(\sqrt{1+x} + \sqrt{1-x}) - \frac{1}{2} \int \left(\frac{1}{\sqrt{1-x^2}} - 1 \right) dx = x \ln(\sqrt{1+x} + \sqrt{1-x}) - \frac{1}{2} \arcsin x + \frac{x}{2} + C.$$

6. $\int_0^1 \frac{\arcsin \sqrt{x}}{\sqrt{1-x(1-x)}} dx$

解: 区间再现—— $I = \int_0^1 \frac{\arcsin \sqrt{x}}{\sqrt{1-x(1-x)}} dx = \int_0^1 \frac{\arcsin \sqrt{1-x}}{\sqrt{1-(1-x)x}} dx = \frac{1}{2} \int_0^1 \frac{\arcsin \sqrt{x} + \arcsin \sqrt{1-x}}{\sqrt{1-x(1-x)}} dx$

$$= \frac{1}{2} \int_0^1 \frac{\frac{\pi}{2}}{\sqrt{x^2-x+1}} dx = \frac{\pi}{4} \int_0^1 \frac{1}{\sqrt{x^2-x+1}} dx = \frac{\pi}{4} \int_0^1 \frac{1}{\sqrt{\left(x-\frac{1}{2}\right)^2 + \frac{3}{4}}} dx$$

$$= \frac{\pi}{4} \ln \left| \left(x - \frac{1}{2}\right) + \sqrt{x^2-x+1} \right| \Big|_0^1 = \frac{\pi}{4} \ln \frac{\frac{1}{2}+1}{-\frac{1}{2}+1} = \frac{\pi}{4} \ln 3.$$

注: 关于恒等式 $\arcsin \sqrt{x} + \arcsin \sqrt{1-x} \equiv \frac{\pi}{2}$ 的证明——

设 $f(x) = \arcsin \sqrt{x} + \arcsin \sqrt{1-x}$, $f'(x) = \frac{1}{\sqrt{1-(\sqrt{x})^2}} \cdot \frac{1}{2\sqrt{x}} - \frac{1}{\sqrt{1-(\sqrt{1-x})^2}} \cdot \frac{1}{2\sqrt{1-x}} = 0$,

故 $f(x)$ 为常函数. 又 $f(0) = \arcsin 0 + \arcsin 1 = \frac{\pi}{2}$, 故 $\arcsin \sqrt{x} + \arcsin \sqrt{1-x} \equiv \frac{\pi}{2}$.

7. 设 $f(x) = \int_{-1}^1 |t-x| e^{-t^2} dt$ ($-1 < x < 1$) 的单调区间与极值.

解: 分区间, 打开绝对值即可. $f(x) = \int_{-1}^1 |t-x| e^{-t^2} dt = \int_{-1}^x (x-t) e^{-t^2} dt + \int_x^1 (t-x) e^{-t^2} dt$

$$= x \int_{-1}^x e^{-t^2} dt - \int_{-1}^x t e^{-t^2} dt + \int_x^1 t e^{-t^2} dt - x \int_x^1 e^{-t^2} dt.$$

故 $f'(x) = \left(1 \cdot \int_{-1}^x e^{-t^2} dt + x e^{-x^2} \right) - x e^{-x^2} - x e^{-x^2} - \left(1 \cdot \int_x^1 e^{-t^2} dt - x e^{-x^2} \right) = \int_{-1}^x e^{-t^2} dt - \int_x^1 e^{-t^2} dt.$

$f''(x) = 2e^{-x^2} > 0$, 故 $f'(x)$ 单调递增. 又 $f'(0) = 0$, 故 $f(x)$ 在 $(-1, 0)$ 单调递减, 在 $(0, 1)$ 单调递增.

故极小值为 $f(0) = \int_{-1}^1 |t| e^{-t^2} dt = 2 \int_0^1 t e^{-t^2} dt = -e^{-t^2} \Big|_0^1 = 1 - \frac{1}{e}$, 无极大值.

8. 设 $f(x) = \int_1^x e^{-t^2} dt$, 求 $\int_0^1 x^2 f(x) dx$.

解: $I = \int_0^1 f(x) d \frac{x^3}{3} = \frac{x^3}{3} f(x) \Big|_0^1 - \int_0^1 \frac{x^3}{3} e^{-x^2} dx = (0-0) - \frac{1}{6} \int_0^1 x^2 e^{-x^2} d(x^2) = -\frac{1}{6} \int_0^1 t e^{-t} dt$

$$= \frac{1}{6} \int_0^1 t d e^{-t} = \frac{1}{6} t e^{-t} \Big|_0^1 - \frac{1}{6} \int_0^1 e^{-t} dt = \frac{1}{6} e^{-1} + \frac{1}{6} e^{-t} \Big|_0^1 = \frac{1}{6} e^{-1} + \frac{1}{6} (e^{-1} - 1) = \frac{1}{3} e^{-1} - \frac{1}{6}$$

9. 设 $f(x) = x\sqrt{2x-x^2} + \int_0^2 xf(x)dx$, 求 $\int_0^1 f(x)dx$.

解: 设 $\int_0^2 xf(x)dx = A$, 在题干等式两端同时乘以 x , 然后在 $[0, 2]$ 积分, 得——

$$A = \int_0^2 x^2 \sqrt{2x-x^2} dx + A \int_0^2 x dx = \int_0^2 x^2 \sqrt{2x-x^2} dx + 2A.$$

$$\int_0^2 x^2 \sqrt{2x-x^2} dx = \int_0^2 x^2 \sqrt{1-(x-1)^2} dx \stackrel{x-1=\sin t}{=} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (1+\sin t)^2 \cos^2 t dt = 2 \int_0^{\frac{\pi}{2}} (1+\sin^2 t) \cos^2 t dt$$

$$= 2 \int_0^{\frac{\pi}{2}} (1+\sin^2 t)(1-\sin^2 t) dt = 2 \int_0^{\frac{\pi}{2}} (1-\sin^4 t) dt = 2 \left(\frac{\pi}{2} - \frac{3}{4} \frac{1}{2} \frac{\pi}{2} \right) = \frac{5}{8} \pi.$$

$$\text{故 } A = \int_0^2 x^2 \sqrt{2x-x^2} dx + A \int_0^2 x dx = \frac{5}{8} \pi + 2A, \text{ 解得 } A = -\frac{5}{8} \pi.$$

$$\text{故 } f(x) = x\sqrt{2x-x^2} - \frac{5}{8} \pi.$$

$$\begin{aligned} \int_0^1 f(x)dx &= \int_0^1 \left[x\sqrt{2x-x^2} - \frac{5}{8} \pi \right] dx = -\frac{5}{8} \pi + \int_0^1 x\sqrt{2x-x^2} dx = -\frac{5}{8} \pi + \int_0^1 x\sqrt{1-(x-1)^2} dx \\ &\stackrel{x-1=\sin t}{=} -\frac{5}{8} \pi + \int_{-\frac{\pi}{2}}^0 (1+\sin t) \cos^2 t dt = -\frac{5}{8} \pi + \frac{1}{2} \frac{\pi}{2} - \frac{1}{3} \cos^3 t \Big|_{-\frac{\pi}{2}}^0 = -\frac{3}{8} \pi - \frac{1}{3}. \end{aligned}$$

10. 设 $f(x)$ 在 $[a, b]$ 上二阶连续可导. 证: 存在 $\xi \in (a, b)$, 使得 $f(b) + f(a) - 2f\left(\frac{a+b}{2}\right) = \frac{(b-a)^2}{4} f''(\xi)$

解: 将端点在中点处展开, 并联合两个等式即可.

$$f(a) = f\left(\frac{a+b}{2}\right) + f'\left(\frac{a+b}{2}\right) \frac{a-b}{2} + \frac{f''(\xi_1)}{2} \left(\frac{a-b}{2}\right)^2, \quad a < \xi_1 < \frac{a+b}{2} \dots \textcircled{1}$$

$$f(b) = f\left(\frac{a+b}{2}\right) + f'\left(\frac{a+b}{2}\right) \frac{b-a}{2} + \frac{f''(\xi_2)}{2} \left(\frac{b-a}{2}\right)^2, \quad \frac{a+b}{2} < \xi_2 < b \dots \textcircled{2}$$

$$\text{将上面两式相加, 得 } f(a) + f(b) - 2f\left(\frac{a+b}{2}\right) = \frac{(b-a)^2}{8} [f''(\xi_1) + f''(\xi_2)] \dots \textcircled{3}.$$

由于 $f''(x)$ 在 $[a, b]$ 连续, 故由闭区间上连续函数的介值定理可得,

$$\text{存在 } \xi \in (\xi_1, \xi_2) \subset (a, b), \text{ 使得 } f''(\xi) = \frac{f''(\xi_1) + f''(\xi_2)}{2}, \text{ 将该式代入 } \textcircled{3} \text{ 中可得,}$$

$$f(b) + f(a) - 2f\left(\frac{a+b}{2}\right) = \frac{(b-a)^2}{4} f''(\xi), \text{ 证毕!}$$